

A Framework for the Structured Implementation of Process Mining for Audit Tasks

Tassilo L. Föhr^{a,*}, Valentin Reichelt^a, Kai-Uwe Marten^a, Marc Eulerich^b

^a Ulm University, Germany

^b University of Duisburg-Essen, Germany

ARTICLE INFO

JEL Classifications:

M40
M41
M42
M48

Keywords:

Process Mining
Internal Controls
Auditing Standards
Design Science

ABSTRACT

Process Mining (PM) enhances the evaluation of the internal control system with corresponding tests of controls due to a comprehensive analysis of variants within business processes. It can be used by external and internal auditors to improve audit efficiency, effectiveness, and quality. Nevertheless, PM is still not an industry-wide best-practice standard, especially due to existing implementation barriers for practitioners. This study utilizes Design Science Research (DSR) to develop the Audit Process Mining (APM) Framework to implement PM into audit tasks and overcome existing implementation barriers. The APM Framework was designed using empirical insights extracted from interviews with subject matter experts across various audit firms and large industrial companies. Furthermore, 19 auditing professionals confirmed that the APM Framework is a valid and verified solution.

1. Introduction

Auditing and accounting procedures are currently evolving due to the incorporation of big data analytics and related emerging technologies (Issa et al., 2016; Kokina and Davenport, 2017). Large amounts of financial data stored in information systems build the basis for advanced data analytics (Appelbaum et al., 2018; Appelbaum et al., 2017; Alles et al., 2021; Austin et al., 2021; Eilifsen et al., 2020; Gepp et al., 2018; Salijeni et al., 2019). This digital transformation is causing a paradigm shift in the auditing profession for both internal and external auditors, as emerging technologies can potentially create value for decision makers and give them a deeper business understanding (Appelbaum et al., 2018; Christ et al., 2021a; Eulerich et al., 2022; Sun, 2019). Auditors also apply advanced data analytics to create added value for their stakeholders by proposing new analytical capabilities, as well as improved audit quality (Manita et al., 2020).

Additionally, visualizations of data and processes through dashboards are helpful in assessing risks, planning audit procedures, and discussing generated insights with clients and stakeholders (Buchheit et al., 2020; Dilla and Raschke, 2015; Salijeni et al., 2021). Information technologies and advanced data analytics are important tools for the automation of routine tasks and therefore enable auditors to invest more time in knowledge-intensive work (Eulerich et al., 2022). Therefore, auditors integrate digital tools into their IT strategies (Krieger et al., 2021; Lowe et al., 2018).

Process Mining (PM) is especially useful for increased effectiveness and efficiency of audit procedures to obtain audit evidence (Eulerich et al., 2023; Jans and Eulerich, 2022; Werner et al., 2021). PM derives and visualizes business processes based on event logs,

* Corresponding author.

E-mail address: tassilo.foehr@uni-ulm.de (T.L. Föhr).

which are stored in various data tables of information systems and belong to certain business cases (van der Aalst, 2016). It offers multifaceted advantages and is primarily characterized by its ability to identify discrepancies and potential internal control weaknesses within business processes by comparing the visualized process flow with desired business process sequences (Davenport and Spanyi, 2019; EY, 2019; Grisold et al., 2021; van der Aalst, 2011, 2012; van der Aalst, 2016). Such visualized processes help mitigate risks and inefficiencies by identifying material deviations (Davenport and Spanyi, 2019; Park and van der Aalst, 2020; van der Aalst, 2016; vom Brocke et al., 2021). When auditing, PM can enhance the evaluation of the internal control system (International Standard on Auditing (ISA) 315 (Revised 2019), IAASB, 2019) and test of controls (ISA 330, IFAC, 2009f) by identifying internal control violations, such as missing approval activities and segregation of duty conflicts (Chiu and Jans, 2019; Jans et al., 2014), or automatically analyzing the three-way match (Werner et al., 2021). PM aids in substantive analytical procedures (ISA 520, IFAC, 2009h) and in the various selection of processes that inherit suspicious patterns (Werner and Gehrke, 2019). PM also enhances responses to identified risks (ISA 330, IFAC, 2009f) by detecting the activities from a derived process model that does not comply with specific accounting standards (e.g., revenue recognition standards) (Wang et al., 2020).

However, analyzing the existing literature and empirical insights from interviews with 14 subject matter experts allowed us to identify some implementation barriers preventing the widespread adoption of PM among auditors. Thus, it is still not clear why and how PM should be the technology of choice for auditors. Moreover, there is no audit-specific implementation guide about what successful PM implementation would look like.

This paper fills this gap and utilizes Design Science Research (DSR) (Geerts, 2011; Peffers et al., 2007) to develop the *Audit Process Mining (APM) Framework*, which provides guidance for implementing PM for auditing. The *APM Framework* serves as an anchor with which an organization can structure the implementation and application of PM for several audit tasks. It was demonstrated to 14 subject matter experts and evaluated by 19 PM experts to ensure its functionality for the profession.

To the best of our knowledge, the *APM Framework* is the first rigorously developed and evaluated guide to structure the implementation of PM into audit tasks based on the following phases: (0) Preliminary phase, (1) Business process understanding and planning, (2) Event data understanding and extraction, (3) Data model and event logs creation, (4) PM and analysis, (5) Audit objectives evaluation, and (6) Deployment, maintenance, and improvement.

Therefore, the contributions of this study are as follows: First, we present an innovative framework to overcome existing implementation barriers associated with PM. Additionally, we empirically demonstrate and evaluate the *APM Framework* to a representative sample of PM experts to ensure the improvements associated with this artifact.

The study proceeds as follows. After exploring related work about PM in auditing, we detail our research approach. In section four, we describe the design and development of our artifact. Section five demonstrates our *APM Framework*, after which we evaluate this demonstrated artifact in section six. Section seven details the limitations of the study and future research opportunities based on our research. The paper ends with a conclusion.

2. Related work

We analyzed prior literature¹ to search for already existing frameworks regarding the implementation of PM for auditing. In general, audit firms and clients implement more data analytics tools in an increased digital accounting ecosystem (Alles et al., 2021; Austin et al., 2021; Dagilienė and Klovienė, 2019; IAASB, 2021). PM as a data analytics technology is relatively new to auditing but reached attention in research and practice (IAASB, 2021; Jans et al., 2013; Jans and Eulerich, 2022; Werner, 2019; Werner, 2017; Werner et al., 2021). Accounting and other related business processes generate relevant data for applying PM in auditing (Amani and Fadlalla, 2017; Jans and Eulerich, 2022). One of the first papers on PM in auditing is the study of van der Aalst et al. (2010) providing an overview of potential applications and challenges.

Early papers exploring PM in auditing provide an overview of potential application areas and associated implementation challenges (Jans et al., 2014; Jans et al., 2013; van der Aalst et al., 2010; Werner and Gehrke, 2015). Later research shows how PM enables a deeper understanding of business processes based on event logs (Chiu et al., 2019; Werner, 2017) and how PM can aid auditors during a financial statement audit (Chiu et al., 2019; Jans and Eulerich, 2022; Werner, 2019; Werner et al., 2021). Most research contributions focus on highly specialized use cases for PM, such as fraud detection (Baader and Krcmar, 2018; Chiu et al., 2020; Jans et al., 2011; Sarno et al., 2020), internal control testing (Chiu and Jans, 2019; Chiu et al., 2019; Eulerich et al., 2023; Werner and Gehrke, 2019), journal entry testing (Stephan et al., 2021), IT systems audits (Zerbino et al., 2018), and the alignment of business processes to specific accounting standards (Wang et al., 2020). Existing research also illustrates some use cases for internal auditing procedures (e.g., Christ et al., 2021b; Jans and Eulerich, 2022; Celonis, 2017; Eulerich et al., 2023) and combinations with other emerging technologies such as Artificial Intelligence, Machine Learning or Robotic Process Automation (Jans and Hosseinpour, 2019; Leno et al., 2021).

Most of the existing research applies PM in case or field studies within single audit firms or companies. These results may not translate to every audit firm or Internal Audit Function (IAF), as the specific conditions and tone of the top can differ significantly within audit firms or IAFs. Furthermore, little guidance is provided for implementing PM as an audit tool. Werner et al. (2021)

¹ We analyzed relevant journals in the American Accounting Association digital library and other relevant sources, like International Journal of Accounting Information Systems and Contemporary Accounting Research. Furthermore, we also used Google Scholar, the world's largest academic database, to ensure that further scientific contributions in the investigated area were covered and to include conference papers etc. that are not in other databases. We filtered the search results by title, abstract, and keywords to identify relevant literature involving "process mining" and "auditing".

presented the initial guidelines regarding how to implement PM during a financial statement audit. The authors referenced their guidance to the appropriate ISAs and focused on PM use cases instead of providing overarching implementation guidelines for PM in risk-based auditing.

Therefore, we identified, from a research-based perspective, that there is a lack of overarching PM implementation guidance among supporting auditors. This prevents the profession from implementation failures (Grisold et al., 2021; Mans et al., 2013; Martin et al., 2021; Thiede et al., 2017; vom Brocke et al., 2021) and fosters the need for more guidance regarding how to implement PM during audits.

3. Research approach

The study at hand follows DSR, which is focused on rigorously developing and evaluating an innovative solution to an existing real-world problem or improving an existing solution using the superior functionality of an artifact (Gregor and Hevner, 2013; Hevner et al., 2004). Our major objective is to enable auditing professionals to overcome existing implementation barriers for a structured implementation of PM in audit tasks. We decided to base the development of our central artifact on several generate-test cycles. This means developing some design specifications and directly testing them against previously defined criteria (Simon, 1996). The research approach we followed was defined by Peffers et al. (2007) and proceeds as follows:

We analyzed existing literature (see Section II) and identified the need for frameworks with structured PM implementation guidelines for auditing.² Furthermore, we analyzed answers from interviews with 14 subject matter experts³ (see Appendix I) about their current experiences with PM, relevant socio-technical factors, and most important challenges for its implementation.

To identify appropriate PM experts, we used our professional network (Power and Gendron, 2015). To avoid potential bias, we interviewed different experts from different auditing firms (including Big Four and Next Ten) and internal audit functions from large industrial companies via online video conferences (see Table 1). The interviews in this study were done in two stages. The first stage was to obtain preliminary insights to identify the study's problem. The second stage of the interviews consisted of questions related to each phase of the developed *APM Framework* to gather some empirical evidence. Overall, the interviews lasted around 60 min and were attended and conducted by (at least) two co-authors.

Answers to the questions asked in the first stage of the interviews indicate that PM projects are still in their pilot testing phases and therefore not entirely embedded and deployed into existing audit methodologies. There are major challenges regarding data understanding and extraction due to the incompatibility of clients' information systems and existing PM tools. Although there are some benefits due to the implementation of PM, much effort is required to prepare data and to investigate mined process variants. There are also major concerns about data privacy, communication with the client, and the complexity of the PM results. These empirically gathered insights indicate that there is a lack of implementation guidelines to overcome existing challenges and to reap large benefits from PM.

Based on these insights, we defined more concrete design objectives to overcome these existing implementation barriers and to enable widespread adoption of PM within the auditing profession. The first objective is to overcome existing PM implementation barriers by using guidelines that explain how to implement PM for auditing within organizations. Such a framework overcomes the existing implementation barriers that prohibit the widespread adoption of PM within the profession. Furthermore, we defined a second design objective to provide this guidance in the form of a framework with clearly distinguishable phases that auditors can follow.

For the design and development of this framework, we relied on an iterative process based on existing theoretical frameworks, such as the hierarchical Cross Industry Standard Process for Data Mining (CRISP-DM) (Chapman et al., 1999) and the PM² framework (van Eck et al., 2015) to ensure a rigorous development of our artifact in accordance with Hevner et al. (2004) and Simon (1996). We utilized many iterations and also implemented existing auditing standards from the external and internal auditing profession. The whole design and development process is described in the next section.

We demonstrated the *APM Framework* during the main stage of our interviews with the previously mentioned 14 subject matter experts to gain empirical insights for each phase of our framework. To develop the main part of our interview script, we followed the recommendations from Miles and Huberman (1994) and used the developed phases of the *APM Framework* to structure the interviews. We presented and described each phase of the *APM Framework* to each interviewee via screen sharing. We conducted interviews until additional interviews became redundant (Lincoln and Guba, 1985). We demonstrated the figure of our conceptually developed *APM Framework* (see Fig. 3) to provide the experts with basic knowledge about the developed artifact. Afterward, we demonstrated and explained each phase of the framework in detail and asked the experts for insights regarding each phase of the *APM Framework* phases. We implemented one empty column for empirical knowledge and insights of each developed phase within a data collection template (see Appendix II) to take notes and collect the empirical data. The interviewees were able to see the notes we took about each phase of

² There are some initial approaches on how to apply PM for auditing (Chiu and Jans, 2019; Jans et al., 2014; Werner et al., 2021). However, these contributions do not focus on a holistic approach.

³ The interviews conducted in this study were based on two stages. The first stage was to obtain preliminary insights to identify our problem. The second stage builds the main part and helps us to demonstrate the *APM Framework*.

TABLE 1
Interview Partners.

Expert	Firm	Focus	Position
1	Second Tier	Audit Data Analytics	Senior Manager
2	Second Tier	Internal Audit	Director
3	Second Tier	Audit Data Analytics	Partner
4	Second Tier	Audit Data Analytics	Partner
5	University	Accounting and Data Analytics	Professor
6	Second Tier	Audit Data Analytics	Partner
7	Second Tier	IT-Audit	Senior Manager
8	Big Four	Audit Data Analytics	Partner
9	Big Four	Data Analytics	Senior Manager
10	Big Four	Audit Data Analytics	Partner
11	Big Four	Audit Data Analytics	Senior Manager
12	Industry	Internal Audit	Internal Auditor
13	Industry	Internal Audit	Internal Auditor
14	Industry	Internal Audit	Internal Auditor

the *APM Framework* at any time through the split screen. For our in-depth analysis of our interview data, we aggregated recurring empirical knowledge and insights for each phase of our *APM Framework* to define the process instances of each main phase in a master data file.⁴ After each session of expert interviews, we evaluated components to meet source credibility criteria according to [Schaefer and Alvesson \(2020\)](#). Thus, we ensure that the gathered empirical knowledge was trustworthy.

For the evaluation of our developed and demonstrated *APM Framework*, we measured its overall usefulness with 19 auditing professionals based on established DSR evaluation criteria ([Hevner et al., 2004](#); [March and Smith, 1995](#); [Sonnenberg and vom Brocke, 2012](#); [Venable et al., 2012](#)). This ensures that our central artifact has direct implications for the auditing profession ([Burton et al., 2021](#); [Peffer et al., 2018](#); [Summers and Wood, 2017](#); [Waymire, 2012](#)).

4. Design and development

The development process used to achieve our previously defined design objectives was based on many generate-test cycles in alignment with [Hevner et al. \(2004\)](#) and [Simon \(1996\)](#). As illustrated in [Fig. 1](#), we chose a multi-methodological approach based on [Nickerson et al., \(2013\)](#) and [Berger et al. \(2020\)](#), which distinguishes the development between conceptual-to-empirical and empirical-to-conceptual iterations. The deductive conceptual-to-empirical development iterations are based on already existing guidelines for PM without the consideration of empirically gathered data (development iteration 1–3). The inductive empirical-to-conceptual development iteration is based on a comprehensive study with subject matter experts to illustrate empirically gathered process instances from the real-world for each phase of our *APM Framework* and to further inform the development of our central artifact (development iteration 4) ([Nickerson et al., 2013](#)).

We initially based the guidance provided within our *APM Framework* on the CRISP-DM process model ([Chapman et al., 1999](#)) and focused it on PM within an audit environment. CRISP-DM is an established standard for data mining and is suitable for PM projects due to its general character covering a broad range of emerging technologies. It defines the implementation guidance for emerging technologies at four different levels of abstraction: (1) phases, which describe the main anchors to rely on during implementation; (2) generic tasks, which further define each phase with generic subtasks relevant for a broad range of emerging technologies; (3) specialized tasks to address how to execute the generic tasks within specific situations (e.g. using PM to perform auditing); and (4) process instance to show implementation projects based on guidance out of the previous levels. Thus, the level of the process instance represents what happens during specific implementation projects.

Following these four levels of abstractions from CRISP-DM the development of our *APM Framework* can be described as follows ([Fig. 2](#)):

The main phases of our *APM Framework* (Level 1) are as follows: (0) Preliminary phase, (1) Business process understanding and planning, (2) Event data understanding and extraction, (3) Data model and event logs creation, (4) PM and analysis, (5) Audit objective evaluation, and (6) Deployment, maintenance, and improvement.

We used components of the PM² framework as stated by [van Eck et al. \(2015\)](#) to define generic tasks for each main phase of the *APM Framework* (conceptual-to-empirical) and further detail our artifact by tasks related to PM for audits with information about input,

⁴ We explicitly refrained from coding for our in-depth analysis of interview data. The individual phases of the *APM Framework* already implicitly represented the underlying codes (each main phase represents one code). Each phase of the *APM Framework* was further divided by subcodes (input, output, resources, activity). Therefore, the answers of the interviewees were already aligned to the right codes and subcodes during the interview by taking notes in the appropriate space of the data collection template.

Iteration	1	2	3	4
Approach	conceptual-to-empirical	conceptual-to-empirical	conceptual-to-empirical	empirical-to-conceptual
Research	CRISP-DM	PM ²	ISA and IPPF AS/PS	Expert Interviews

Fig. 1. Research Procedure (based on Berger et al. 2020).

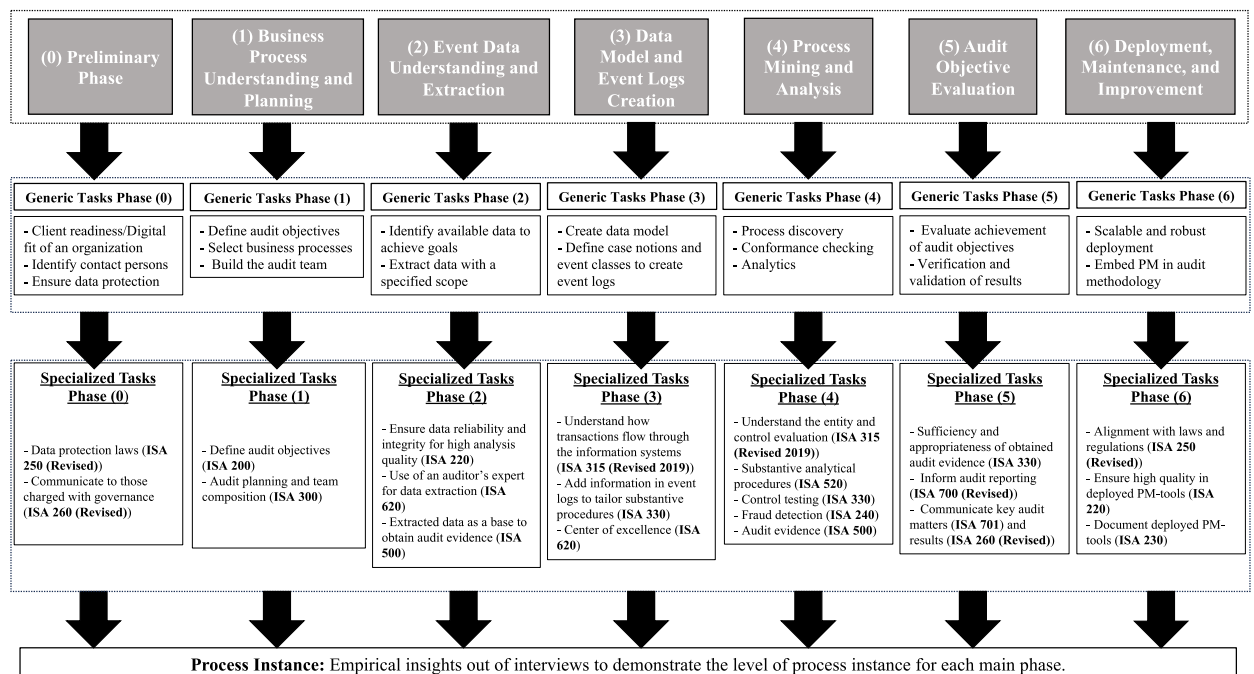


Fig. 2. Framework Layers of Abstraction.

output, resources, activities, and team compositions.

To define the next layer of abstraction within our *APM Framework* (Level 3), we aligned the previously defined generic tasks to the regulatory and legal framework of the risk-based audit approach defined by the ISAs, the attribute standards, and performance standards of the International Professional Practices Framework (IPPF).⁵ This specialized level of abstraction ensures that the *APM Framework* gives specialized guidance for auditors regarding how to implement PM to conduct audit tasks in alignment with the

⁵ We name all the internal auditing standards from the Institute of Internal Auditors (IIA) according to the International Standards for the Professional Practice of Internal Auditing (Standards). For more information see IIA (2016). These are based on the International Professional Practices Framework (IPPF). For more information see IIA (2017). In the manuscript, we name the abbreviation of the standards IPPF AS (Attribute Standards) and IPPF PS (Performance Standards) with the specific numbers we referenced to.

relevant auditing standards.⁶

To define the process instance layer of our *APM Framework* (Level 4), we gathered empirical insights from the previously described interviews with 14 subject matter experts (Rowley, 2012) and thus extended our conceptually-based *APM Framework* from the previous iterations with empirical measured experiences (empirical-to-conceptual). These insights give detailed information about how each phase of our *APM Framework* is executed or not within a real-world setting. This conducted survey is an essential component of the *APM Framework*'s development and demonstration process, as some empirical insights extend our *APM Framework*. An example of this was adding a phase 0, defined as a preliminary phase, which was suggested by some of the subject matter experts. We explicitly demonstrated and explained each phase of the *APM Framework* to the experts to gather empirical insights regarding how and if the framework can overcome some of the existing implementation barriers and meet the overall objectives.

Since the demonstration is about to show "the use of the artifact to solve one or more instances of the problem" (Peffer et al., 2007, p. 55), the last process instance layer of development (Level 4) can be also seen as a comprehensive demonstration of the artifact. We demonstrated 14 instances of how the *APM Framework* overcomes existing implementation barriers in this study by interviewing 14 subject matter experts regarding their perceptions about the *APM Framework*.

Following Gregor and Hevner (2013), our developed *APM Framework* extends the basic frameworks of CRISP-DM (Chapman et al., 1999) and PM² (van Eck et al., 2015) and therefore provides a central instance for organizations with specialized guidelines to be adopted by auditing professionals in order to successfully implement PM. CRISP-DM is a general data mining process model with no special focus on PM. The framework of van Eck et al. (2015) comes with a detailed focus on PM, but it does not provide any specialized auditing guidelines, such as the alignment of the frameworks with existing ISAs or IPPF AS and IPPF PS. Therefore, the *APM Framework* extended the initial guidance provided by Chapman et al. (1999) and van Eck et al. (2015) with a special focus on the implementation of PM in auditing for each of the developed main phases. Thus, the *APM Framework* represents overarching guidance especially relevant for auditing professionals aligned toward existing auditing standards (e.g., ISAs). Furthermore, each phase of our *APM Framework* is detailed by process instances gathered from interviews with auditors. This extends to CRISP-DM and PM², as their initial development was not based on empirical gathered data obtained from interviews.

5. The audit process Mining Framework

5.1. Generic View

In this section, we demonstrate that the *APM Framework* is the core of our research. We first give a broad overview of the framework and then demonstrate the framework based on its four different layers of abstraction. Therefore, we demonstrate the *APM Framework* based on its main phases, generic tasks with a focus on PM, and auditing standards to assure specialized audit guidance. Each phase is enriched with empirical insights in the last layer to show how PM projects are concretely executed within audit engagements.

The *APM Framework* guides auditors during the implementation of PM to conduct audit procedures. The main audit objective associated with PM can be broadly defined as more advanced internal control evaluation (ISA 315 (Revised 19), IAASB, 2019; IPPF PS 2130, IIA, 2016) and enhanced skills for test of controls (ISA 330, IFAC 2009f; IPPF PS 2320, IIA, 2016). Implementing PM in alignment with the *APM Framework*, helps achieve our objectives. An overview of the developed *APM Framework* is shown in Fig. 3.

Its seven phases are defined by different objectives, described as goal- and data-related objects, and results. Goal-related objects are described as (1) defined audit objectives, and (2) audit evidence obtained from audit procedures conducted with PM. Data-related objects are defined by the several components necessary to compose the relevant data to utilize PM: (1) information systems as the main basic component to extract relevant and process-related (2) event data, which should be aligned to specific and discrete activities, (3) data models to define relations of event data, and (4) event logs as the basic component for PM analysis as they inherit case notions and event classes. In our results, we defined mined (1) process models and (2) conducted analytical audit procedures based on the mined process models. Process models correspond to the ordering of activities related to specific business processes and illustrate various process variants within an organization, whereas analytical audit procedures are defined as audit tasks to obtain audit evidence from the mined process models and corresponding variants (van Eck et al., 2015). Table 2 illustrates the main content of the *APM Framework*.

⁶ For reasons of simplicity, we only referenced to the ISAs in Fig. 2. However, the main section of this paper also includes the IPPF AS/PS.

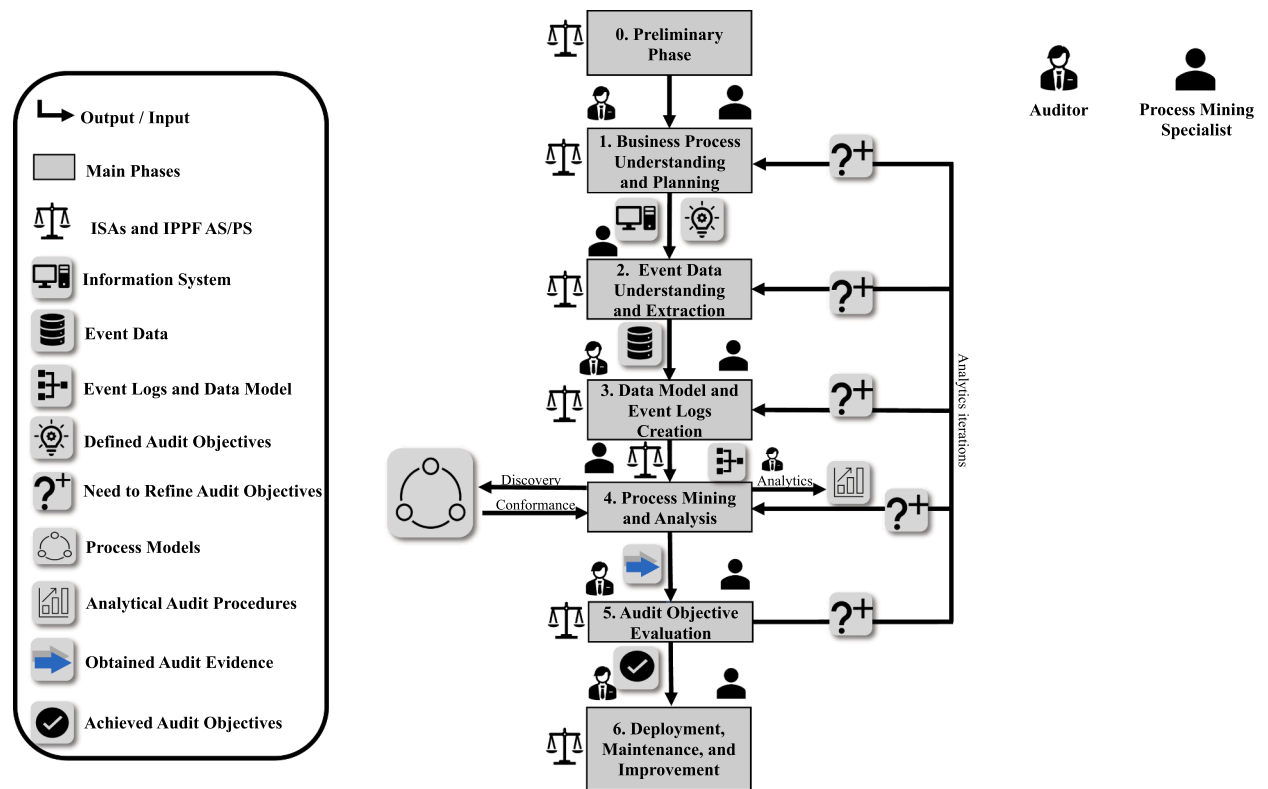


Fig. 3. APM Framework (based on Chapman et al. 1999; van Eck et al. 2015).

5.2. Preliminary phase

Following the recommendations of the surveyed experts, a Phase 0 was added to the *APM Framework*. Therefore, before the implementation of PM, auditors should clarify the client's or IAF's readiness in advance and identify appropriate contacts within the organization to ensure efficient data extraction and process knowledge for business processes. This can be achieved if the auditor communicates with those in charge of governance (ISA 250 (Revised), IAASB, 2016; IPPF PS 2020, IIA, 2016). If necessary, the audit team and the client's or IAF's management need to consult with work councils to discuss data protection and other concerns to ensure the adherence of PM projects to existing laws and regulations (ISA 260 (Revised), IAASB 2015a; IPPF PS 2110, IIA, 2016).

TABLE 2
Content of APM Framework.

Phase	Activities
Preliminary Phase (0)	1. Check client readiness/digital fit of an organization 2. Identify contact persons within the company for data extraction and sufficient process knowledge 3. Discuss data protection issues
Business Process Understanding and Planning (1)	4. Define the audit objectives 5. Select the relevant business processes 6. Build the audit team
Event Data Understanding and Extraction (2)	7. Identify and understand available data to achieve audit objectives 8. Integrate different data sources from various information systems to define and extract event data 9. Transfer existing process knowledge
Data Model and Event Logs Creation (3)	10. Create data model to define relations between extracted data 11. Create event logs by defining case notions and event classes based on event data
Process Mining and Analysis (4)	12. Process discovery 13. Conformance checking
Audit Objective Evaluation (5)	14. Analytical audit procedures 15. Assess the extent to which the audit objectives can be achieved with PM 16. Verify and validate results
Deployment, Maintenance, and Improvement (6)	17. Evaluate if further iterations between phases 1–4 are necessary 18. Deploy PM if audit objectives can be achieved 19. Embed and scale PM into the existing audit methodology 20. Implement monitoring and maintenance mechanisms to periodically review PM functionalities

5.3. Business Process Understanding and Planning

5.3.1. Define Audit Objectives

The objective of an auditor is to obtain reasonable assurance about whether the financial statements are free from material misstatements (ISA 200, [IFAC, 2009a](#)) or to provide an independent and objective assurance service to add value and improve an organization's operations in accordance with relevant auditing standards, regulations, and laws (IPPF AS 1000, [IIA, 2016](#)). Afterward, auditors need to state their findings in an audit opinion within an audit report (ISA 200.11, [IFAC, 2009a](#); IPPF PS 2400 and IPPF PS 2450, [IIA, 2016](#)). The overall success criteria for this phase are the obtained audit evidence attributable to the implementation of PM.

The goal of PM at this stage is mainly driven by checking the conformance of existing business processes to certain business rules and regulations. Therefore, the auditor needs to evaluate the internal control system of an organization with the functionalities of PM (ISA 315.1 (Revised 2019), [IAASB, 2019](#); IPPF PS 2130, [IIA, 2016](#)). The IAF supports the management by detecting certain risks (IPPF PS 2120, [IIA, 2016](#)) and ensuring operational performance and process governance ([Jans and Eulerich, 2022](#)). In sum, PM success criteria are determined through the degree of accuracy with which compliance violations in the operation of internal controls are detected.

Empirical Insights: According to the experts, PM merely helps the client gain a better understanding of their business model and its internal control system through the visualization of all process flows and controls. The starting point of the analysis for external auditors is based on the content of the financial statements. It should be focused on which accounts are fed with which processes. The analysis then focuses on the process level to see whether the monetary values contained in the mined business process correspond to the overall values of the involved accounts connected to the business process in the financial statements.

Analyzing internal controls is a crucial part of the audit process due to mass transactions and big data within certain business models. Therefore, PM improves risk assessment by using the generated insights as input for the assessment of business and fraud risk as well as internal control deviations. The process model presents all process flows and controls within process variants. Since visualized processes allow risks to be identified based on mined business processes, auditors meet regulatory requirements regarding ISA 315 (Revised 2019), [IAASB, 2019](#). This approach allows us to replace interviews and manual walkthroughs and offers a holistic and objective view of real processes to derive an appropriate audit strategy. This leads to greater efficiency and reduces sample sizes within selected business processes. Moreover, auditors can provide added value for the top management and present visualized business processes for further analytical capabilities using PM results. Making a distinction between existing clients and new clients is recommended when it comes to project planning. For new clients, PM can significantly increase process understanding prior to the actual audit. For existing clients, PM supports the validation of current process understanding and helps to identify controls that the client themselves would not understand. Moreover, existing PM projects can be used to apply more data analyses based on the extracted data.

Internal auditing focuses more on process governance like analyzing whether processes are compliant with the company's internal guidelines or performed efficiently without relevant risks. A purely exploratory research design is difficult for internal auditors to justify. Therefore, it makes sense to investigate the business processes of the entity based on special auditing questions, which involves using evidence to derive appropriate findings from PM. For example, one can find which transactions have been booked by an administrator and which have not.

5.3.2. Audit Planning

The auditor needs to plan the implementation of PM for an audit to ensure it performs effectively (ISA 300.4, [IFAC 2009d](#); IPPF PS 2010, [IIA, 2016](#); [Jans and Eulerich, 2022](#)). The PM-related audit objectives are based on the business processes to be analyzed during an audit. Therefore, relevant processes need to be identified and the corresponding information systems that store the relevant data need to be selected ([van Eck et al., 2015](#)). Business processes of interest depend on an auditor's choice and the clients' or organizations' circumstances ([Jans, 2019](#)). The auditor should focus on selecting business processes associated with a high probability of material control deviations due to error or fraud (ISA 240.10, [IFAC, 2009c](#); IPPF AS 1210.A1 and A2, [IIA, 2016](#)) or the absence of implemented and effective control mechanisms (ISA 330, [IFAC, 2009f](#); IPPF PS 2130, [IIA, 2016](#)).

Another activity within this planning task is the composition of the audit team. Team members need both accounting knowledge and process understanding. The involvement of different PM and business experts for the assessment of the design and effectiveness of implemented internal controls is valuable (ISA 300.A2, [IFAC, 2009d](#); IPPF PS 2200, [IIA, 2016](#); [Jans and Eulerich, 2022](#)).

Empirical Insights: In an audit context, high-risk processes are in the focus area to be mined by PM. Order-to-Cash, Purchase-to-Pay and several accounting processes are considered relevant by most of the experts within PM projects. These processes are high in volume, highly standardized, and end-to-end embedded within their underlying information systems. These business processes also have a material impact on key balance sheet items. Additionally, auditors determine business processes to be analyzed through PM by considering which ones enhance effectiveness and efficiency the most in audit procedures for external auditors. Regarding the client side, it is important to have identified the process owner, process experts and to have technical support by the clients or the audit firms' IT-department.

Internal auditors merely focus on enhanced compliance with internal process guidelines. Long-term experiences with PM projects have shown, that the inclusion of PM or data analytics experts from the audit firm is beneficial for the whole project. The audit team is supported or educated with technical PM skills because auditors are not capable of building data models in advance even though they usually have the relevant process knowledge.

5.4. Event Data Understanding and Extraction

At the beginning of an audit, there is some iteration between the defined audit objectives and available event data stored in the information systems. Without appropriate and sufficient data, initially specified audit objectives for PM need to be refined to match available event data (Kelleher and Tierney, 2018). For PM purposes, different data sources from various information systems are extracted and joined into one data table containing the event data. Event data can be defined as data without any event classes or case notions. This extracted data builds the base on which audit evidence is obtained with PM (ISA 500, IFAC, 2009g; IPPF PS 2320, IIA, 2016). To extract event data, the auditor needs to define its scope of extraction regarding granularity, period, data attributes, and correlation between event data elements (Jans, 2019). The use of an auditor's expert may be considered due to the technical complexity in this phase (ISA 620, IFAC, 2009j; IPPS AS 1210.A1 and 1220.A1, IIA, 2016). It is important to have technical expertise for data extraction to ensure data reliability and integrity and therefore a high PM analysis quality (ISA 220 (Revised); IAASB, 2020; IPPF AS 1300, IIA, 2016). Another important aspect of this phase is to transfer existing process knowledge simultaneously with the event data extraction from business experts and process analysts through interviews or brainstorming sessions (van Eck et al., 2015).

Empirical Insights: The extraction of data tables from underlying information systems can be done in cooperation with the audit data analytics team or by the IT department of the client itself in consultation with the corresponding audit team. Here, established PM vendors provide requirement documents for the data export. Some audit firms use off-the-shelf solutions and third-party provider, such as Celonis or consulting start-ups for PM, as this facilitates data extraction through specifications (requirements catalog). In some cases, experts use self-developed solutions for data export and data preparation. For a complete data export, auditors must closely cooperate with the client regarding the data tables and feature attributes to be exported and regarding the identification of all relevant subsystems. Data extraction is an iterative process until all required data is available. Basic knowledge of the underlying information system is helpful for auditors. Feature attribute designations can differ between the underlying information system and the PM tool. The application of special data extraction software tools, like SAP, is beneficial. Due to the defined PM and audit goals, data covers one fiscal year or a specific period of interest. In contrast to PM projects from internal auditing, no software may be installed in the scope of an external audit to ensure the independence of the auditor. Some clients have personnel capacities within the IT or internal audit department for data export, but do not accept any additional effort caused by a PM project initialized by the financial statement auditor. During audit planning, this should be considered by the lead engagement partner. Developing and implementing standard interfaces for data extraction is important for software vendors and standard setters who wish to improve future PM projects within financial statement audits.

5.5. Data Model and Event Logs Creation

The data model establishes the structure in which data are stored in an information system, including the relationships between different types of data. Therefore, it represents a metamodel regarding how data is organized and stored in an information system defining various entities (such as customers, orders, and products) and the relationship between these entities. Data models help an auditor understand how event data is processed by the information systems and give some valuable insights into the relationships of entities underlying the business processes to be analyzed (ISA 315 (Revised 2019), IAASB, 2019; IPPF PS 2110 and 2120, IIA, 2016).

The basic data format to mine business processes with PM is an event log, which can be derived with the help of a data model. They can be defined as "specific views on event data, created by defining case notions and event classes" (van Eck et al., 2015, p. 302). A case notion is associated with the corresponding events, which are aggregated and illustrated as a process instance. Event classes focus on different activities within a process instance. In addition, auditors may use different aggregation methods to reduce complexity or add further information to the event log. Furthermore, filtering techniques such as slice or dice, variance-based filtering, or compliance-based filtering can be used (van Eck et al., 2015). These techniques, e.g. adding more information to the event logs, thus tailoring the features of the event logs to the audit procedures to be conducted (ISA 330, IFAC, 2009f; IPPF PS 2210, IIA, 2016), impact the quality of the results. Creating a specialized PM team in a center of excellence with an auditor's experts (ISA 620, IFAC, 2009j; IPPS AS 1210 and 1220, IIA, 2016) can be helpful in scaling up these necessary PM activities (Celonis and Fraunhofer Institute, 2022; Jans and Eulerich, 2022).

Event log creation can be conducted on a transactional basis, in which their linkage of process flows is guaranteed processing their corresponding time stamps. Several off-the-shelf PM tools utilize one data model for one specific business process (e.g., Order-to-Cash, Purchase-to-Pay) based on this time-stamp related event log creation. There is also a more audit-specific event log creation mechanism, called financial PM, especially relevant for financial statement auditors (Gehrke, 2010). It builds event logs for process flow analyses based on the booking logic in financial accounting. Financial PM takes a more holistic approach, as the linking of individual event logs for process visualization is conducted via the accounts managed as open items (Gehrke, 2010; Werner, 2017). For example, in a Purchase-to-Pay process, if goods are delivered, an open item is subsequently posted to the goods receipt clearing account. When the company receives the invoice, which is posted to the goods receipt clearing account open item again, the open item is closed accordingly. This establishes the link between the goods receipt and the invoice. By looking at the payment run, this can also be linked together accordingly.

Empirical Insights: This activity is performed by the audit data analytics team or the IT department within the audit firm or IAF. Internal auditors perform this process step themselves without external support, according to interviewed experts. One task within this activity is checking for the completeness and correctness of extracted data by the audit data analytics team or IAF. During data processing, various tools, such as Power BI, can be applied for data preparation and data model building. A developed data model structures extracted data tables and shows how data are related after joining several event data tables from the source systems. Model

building is performed based on the business process to be mined and its underlying information system. Event logs do not initially exist in the source information systems, which makes this phase time-consuming. Consequently, event logs need to be created based on a data model via SQL queries. This process and source code for event log creation is then accurately documented in the case of subsequent peer review engagements. After building the data model, the audit data analytics team defines activities from the identified business process and summarizes or filters the data. The prepared data model is loaded into the PM tool, which then processes the event logs as a base to generate a visualized process model.

5.6. Process Mining and Analysis

This phase focuses on applying PM to the previously generated event logs to meet the defined audit objectives. Therefore, process discovery and conformance checking with subsequent analytical procedures (ISA 520, IFAC, 2009h; IPPF PS 2320, IIA, 2016) are the most relevant analyses for this phase (van Eck et al., 2015).

An auditor focuses on the identification and evaluation of risks regarding material misstatements based on fraud or error on the financial statement and assertion level during process discovery (ISA 315.11 (Revised 2019), IAASB, 2019; IPPF PS 2120, IIA, 2016). The achievement of the audit objective results in the performance of risk assessment procedures based on the mined event logs illustrated as process variants to obtain an understanding of control environments, the entity's risk assessment process and information system, monitoring of internal controls, and control activities performed (ISA 315 sec. 21–26 (Revised 2019), IAASB, 2019).

Based on this gained understanding of an organization's business processes, an auditor can perform control tests (ISA 330, IFAC, 2009f; IPPF PS 2130, IIA, 2016) to check the conformance of mined process variants against prescribed approval mechanisms to obtain audit evidence (ISA 500, IFAC, 2009g; IPPF PS 2320, IIA, 2016) and to evaluate control weaknesses associated with error or fraud (ISA 240, IFAC, 2009c; IPPF AS 1210.A1 and A2, IIA, 2016).

5.6.1. Process Discovery

The process discovery function (van der Aalst, 2016; van der Aalst, 2012) should be applied as a basis for the modeling and analytics stages. Based on event logs and utilizing a discovery technique (e.g. α -algorithm), a Petri Net illustrating the control flow of the event logs is created (van der Aalst et al., 2004).

The discovered process model of a specific business process can be used to understand how the business processes relate to the posted values of the financial accounts. The posted values of the transactions stated in the process model need to be aggregated and compared to the posted values of financial statement accounts (Werner, 2016; Werner et al., 2021). Based on these findings, PM discovery provides support for the auditor to develop an initial risk assessment (ISA 315.7 (Revised 2019), IAASB, 2019; IPPF PS 2120, IIA, 2016) and a preliminary audit strategy (ISA 300.2, IFAC, 2009d; IPPF PS 2200, IIA, 2016). Also, manual postings to financial accounts not covered by the mined process model and its corresponding activities can be extracted. This helps an auditor to identify unexpected or uncommon activities, that are not embedded into the de jure process model (ISA 240.22, IFAC, 2009c; Werner et al., 2021).

Analyzing the process model in terms of uncommon control flow paths and process variants can help to assess inherent audit risk (ISA 315.4 (Revised 2019), IAASB, 2019; IPPF PS 2120, IIA, 2016) and to determine the feasibility of the audit plan for testing internal controls (ISA 330.6, IFAC, 2009f; IPPF PS 2130, IIA, 2016). The auditor can inspect the process model regarding existing control activities, such as approvals by the managers charged with internal compliance and governance (ISA 315.A158 (Revised 2019), IAASB, 2019; IPPF PS 2130, IIA, 2016). The process model allows the process activities and variants conducted by an entity to be illustrated. A process variant can be seen as an activity within the process model with identical control flow paths (Jans et al., 2014). Thus, the process discovery function enhances process documentation (ISA 230.8, IFAC, 2009b; IPPF PS 2330, IIA, 2016).

Empirical Insights: Process discovery gives an auditor a detailed understanding of the client's or organization's business process and of the implemented internal controls. Nevertheless, the mined process model often illustrates too many variants to audit each process variant in the mined process model. Therefore, the auditor needs to apply risk-based audit procedures to analyze these variants in an efficient manner while regarding time and cost constraints. Based on materiality thresholds, the auditor focuses on the most common variants with highest monetary values derived from the process model to assure reasonable audit assurance over a high materiality scope of process variants, analyzing only a percentage of the most common ones. It is often possible to cluster several variants because the most common variants follow the same procedures. The auditor can possibly enhance its documentation capabilities by using the process discovery function. The documentation of the obtained understanding of the entity and its environment is usually time-intensive, especially regarding the extraction of underlying process mechanisms through management inquiries.

In contrast, internal auditors often have good process understanding and therefore rarely use process discovery. However, internal auditors apply process discovery only when handling newly implemented business processes or improving their understanding of current processes.

5.6.2. Conformance Checking

A target process model and the mined event log-based process model, which record real-world execution behavior, are relevant for checking conformance, which simultaneously detects deviations (Rozinat and van der Aalst, 2008; van der Aalst, 2012). To test the operating effectiveness of internal controls, the control risk should be determined by the auditor (ISA 315.34 (Revised 2019), IAASB, 2019; IPPF PS 2120 and 2130, IIA, 2016). For each significant risk, the auditor needs to evaluate whether the existing control is appropriate for mitigating the corresponding risk of material misstatement (ISA 315.26 (Revised 2019), IAASB, 2019; IPPF PS 2120 and 2130, IIA, 2016). For example, a significant risk could be that a specific purchase order was not signed. Therefore, a corresponding

control could be that no purchases can be made without approval indicated by the activity “sign”. The auditor can investigate whether the control flow path of the mined process model includes these stated control activities (Werner et al., 2021).

Different analysis techniques to understand material deviations can be utilized by the auditor regarding a variant analysis. Therefore, the auditor needs to distinguish between standard and nonstandard variants. Non-standard variants can represent inefficient or incorrect business control procedures, whereas standard variants are process deviations within stated business and compliance rules (Chiu and Jans, 2019). Non-standard variants can be categorized as (1) missing activity, (2) activity not in the right order, and (3) redundant activity (Chiu and Jans, 2019). Based on an overall classification of process variants into standard and non-standard variants, the auditor needs to assess the materiality threshold to justify which variants are worthy of further investigation by an auditor (ISA 320.8–9, IFAC, 2009e; IPPF PS 2210, IIA, 2016) and to reduce the number of variants to an adequate amount in accordance with audit efficiency.

Empirical Insights: Conformance checking is an important PM phase for risk-based audit procedures in both, internal and external auditing. During conformance checking, auditors examine process variants that deviate from the happy path or have suspicious loops in the process sequence. Results of detected and potential control deviations need to be validated and verified between the auditor and the process owner from the client side or organization to synthesize relevant responses based on the detected control deviations. On this basis, further analytical audit procedures or interviews can be conducted. For example, analyzing the conformance of event logs based on the Order-to-Cash process can yield different variations. These variations need to be validated by the auditor and process owner. Based on this, further audit procedures can be conducted. A problem in a risk-based audit context could be that it is highly time-intensive to investigate all the detected control deviations in detail. Therefore, the materiality threshold is especially useful in the identification and justification of specific variants for further analysis. Also, specific filtering techniques on high risk-based individual process activities or the detailed investigation of the resource perspective (responsible employee) are relevant audit procedures in this phase. For example, the top five debtors with high monetary value and suspicious process variants can be detailly analyzed through substantive testing procedures. All the other variants (with low monetary value) can be fully investigated by time efficient analytical audit procedures.

Smaller audit clients or organizations do not have any detailed as-is process models and thus do not have a basis to build on for conformance checking analyses. However, most auditors and process owners, especially employees from Public Interest Entities, have a good understanding of the standard process, such as Purchase-to-Pay or Order-to-Cash, and how the specific process activities should be sequentially conducted. Further, there are a lot of implemented control activities in the underlying information systems by default. If there is no explicit process documentation of the as-is process model, then conformance checking can be implicitly done by the auditor.

5.6.3. Process Analytics

PM-related analytical audit procedures (ISA 520, IFAC, 2009h; IPPF AS 1220.A2 and IPPF PS 2320, IIA, 2016) can be conducted by clustering mechanisms and sampling approaches (ISA 530, IFAC, 2009i) of mined process variants based on defined materiality thresholds to obtain audit evidence (ISA 500, IFAC, 2009g; IPPF PS 2320, IIA, 2016) from PM-generated results. Clustering can be concretely used in PM through the comparison of the mined organizational structure with the designed organizational structure from internal compliance rules regarding collusive fraud. For example, if missing activity “sign” in a Purchase-to-Pay process can be identified, a social network analysis can extract different clusters of employees involved into this control violation. Therefore, the employee directly responsible for the control violation build the cluster center and all other employees interacting with this employee are assigned to the same cluster (Jans et al., 2014).

For example, certain underlying controls, such as the activity “sign” and “release”, “release” and “goods receipt” or “goods receipt” and “invoice receipt” of a purchase order, need to be executed by two distinct individuals. In such cases, PM enables the auditor to test operating control effectiveness (ISA 330.4 (b), IFAC, 2009f; IPPF PS 2130, IIA, 2016) using a segregation of duties analysis (Jans et al., 2014). Conducting a segregation of duties analysis and the corresponding identification of control deficiencies incorporates the consideration of allowed client specified exceptions for various circumstances (e.g. holidays, lack of sufficient employees). The auditor can analyze only these employees which have the highest rate in conducting non-standard variations and segregation of duties violations with corresponding high risk (Chiu and Jans, 2019). More detailed insights can be extracted if the auditor conducts a social network analysis showing all the interactions between employees. The highest amount of value can be extracted from this analysis if the auditor focuses on employee subgroups within the company regarding specific materiality thresholds (ISA 320.8, IFAC, 2009e).

In the phase of process analytics, we explicitly leave the analysis capabilities of PM and dive deep into analytical audit procedures (ISA 520, IFAC, 2009h; IPPF PS 2320, IIA, 2016). The aim of the phase of process analytics is to search for specific data features gaining relevant insights to obtain sufficient and appropriate audit evidence (ISA 500.A1, IFAC 2009g; Jans and Eulerich, 2022).

Empirical Insights: Process analytics can be conducted based on the mined results by the application of process discovery and conformance checking to obtain audit evidence. Representative analytical activities can be defined as the clustering, validation, and verification of identified business process deviations per control environment or e.g., per creditor. It is useful for deriving additional audit procedures and sampling procedures based on materiality thresholds. Further, this data driven approach is more objective than manual walkthroughs and interviews with employees.

5.7. Audit Objective Evaluation

The phase of audit objective evaluation serves to assess the extent to which the defined audit objectives (ISA 200.11, IFAC, 2009a; IPPF AS 1000, IIA, 2016) can be achieved by the application of PM. It must be clear to what extent the corresponding analysis approaches can contribute to the fulfillment of the audit and PM success criteria. This means, that an auditor needs to evaluate the sufficiency and appropriateness of obtained audit evidence (ISA 330, IFAC, 2009f; IPPF PS 2320 and 2330, IIA, 2016) to measure achieved audit objectives. The impact of PM for the detection of material misstatements and the degree of accuracy in detecting compliance violations of internal controls with corresponding material weaknesses or misstatements should be clearly stated (Chapman et al., 1999). Therefore, the auditor needs to examine the findings resulting from the conducted audit procedures and the visualized business processes (van Eck et al., 2015). Visualized process models and identified deviations can be discussed with the management and process owners (ISA 260 (Revised), IAASB, 2015a; IPPF PS 2600, IIA, 2016). During the evaluation phase it should be determined if the PM analytical capabilities are sufficient for meeting the success criteria or if another iteration of specific phases of the APM Framework is necessary (Kelleher and Tierney, 2018).

It should be considered which consequences out of the application of PM are drawn from the resulting deviations (ISA 530.12–13, IFAC, 2009i; IPPF PS 2320, IIA, 2016). It may be reasonable to iterate back to the modeling and mining phase (Chapman et al., 1999; van Eck et al., 2015).

Verified and validated findings relevant to achieve defined audit objectives should be utilized to inform audit reporting (ISA 700 (Revised), IAASB, 2015b; IPPF PS 2400, IIA, 2016). Especially, the most relevant findings, such as key audit matters (ISA 701, IAASB, 2015c) or most relevant criteria (IPPF PS 2410, IIA, 2016) should be communicated within this audit report.

Empirical Insights: The auditor should be able to interpret the results correctly and to distinguish unusual or unexpected process flows from expected ones. Also, some verification and validation procedures of the results generated by PM and process analysis should be implemented. This can result in a verification of the findings through the investigation of underlying data sources and system implementations. Validation can be proved through the comparison of unexpected relationships with stakeholders' opinions (i.e., interviewing process owners and responsible management). It is essential to identify the root cause of business process deviations with most material volume. Furthermore, to evaluate the value of the use of PM, the findings from the application of the tool must be clearly attributable to the audit evidence it generated. The most common objective achieved with PM is increased audit efficiency and effectiveness due to more directed audit procedures to obtain audit evidence.

Detailed process documentation provided by the client is helpful for the evaluation of PM generated findings. In some cases, anomalies or missing controls can be clarified by contacting process owners, when e.g., controls are not implemented digitally. These examination steps support deeper process understanding. The remaining findings are analyzed by the conduction of subsequent audit procedures, gathering more information and documents as audit evidence.

So far, there is a lack of clear guidance and instructions for handling deviations in the context of the risk-based audit approach by the standard setters. So, PM is used by the experts from external auditing in an explorative way to obtain audit evidence. The presentation of the PM generated results can be used in benchmarking with subsidiaries to generate recommendations to management and board members as added value resulting from a PM-enhanced audit. Moreover, the application of PM can be a part of workshops offered by the audit firm.

5.8. Deployment, Maintenance, and Improvement

If the application of PM meets its success criteria, the auditor needs to scale the implementation of the PM tool and embed it into existing audit methodologies over several audit engagements. Therefore, the auditor should summarize its deployment strategy regarding the results of the evaluation phase in a deployment plan (Chapman et al., 1999). At this stage, auditors have approved the analytical capabilities of PM, such as process discovery or conformance checking, as meeting the previously defined success criteria. It should be determined, how the PM tool can be implemented into existing audit methodologies and the existing technical infrastructure (Jans and Eulerich, 2022) under the consideration of relevant laws and regulations (ISA 250 (Revised), IAASB, 2016; IPPF PS 2110, IIA, 2016).

It is important, that deployed PM-tools ensure high quality (ISA 220 (Revised), IAASB, 2020; IPPF AS 1300, IIA, 2016) to achieve audit objectives within a live-production environment. Further, the auditor shall implement monitoring and maintenance mechanisms

TABLE 3
Existing Implementation Barriers.

Existing implementation barriers	Votes	Percentage
Challenges in creating the data model	9	47 %
Costs of implementation	7	37 %
Inconsistent data basis	5	26 %
Lack of understanding among users	12	63 %
Lack of use by auditors	9	47 %
No clear use cases within the audit department	7	37 %

Table 4
Survey Results about Perceptions of the APM Framework.

DSR Evaluation Criteria	Mean	Std. Dev	t-stat (t-test against midpoint (0))	Distribution of agreement (−3 to 3)
<i>The APM Framework is useful</i>	1.89	0.94	8.82***	
<i>The APM Framework is easy to use</i>	2.05	0.85	10.55***	

*** Significant at $p < 0.001$, two-tailed t -test. Participants were asked to evaluate their agreement on a 7-point scale. Higher values illustrate that the *APM Framework* meets the assessed DSR evaluation criteria. The t -stat reports on the results of testing whether participants' agreements significantly deviate from the neutral midpoint (0) of the 7-point scale.

to periodically review and document the functionalities of the PM tool (ISA 230, [IFAC, 2009b](#); IPPF PS 2330, [IIA, 2016](#)). The auditor determines if the deployed PM tool still meets the stated success criteria, which may change over time regarding revised or newly announced auditing standards. Also, the data perspective needs to be monitored on a regularly basis. Analyzing a new or adapted business process can result in new analysis approaches making the previous ones obsolete ([Kelleher and Tierney, 2018](#)).

Empirical Insights: PM can either be implemented broadly for all clients/legal entities or specifically for certain industries to answer specific questions or to assess industry risks (e.g. in finance, retail, or public utility companies). Currently, audit firms struggle to develop an appropriate PM strategy to embed the technology into existing audit methodologies. Therefore, several audit firms remain still in an exploratory phase, whereas the implementation for large clients using established information systems is already an ongoing process. Early adopters in internal auditing developed sophisticated PM strategies including detailed training concepts for auditors and increasingly more use cases (e.g. IT audit). In the future, the development of a data lake strategy within the audit firms and a standardized process for data extraction are important for the PM journey. Additionally, guidance by the standard setters and new data interfaces in existing information systems are helpful.

For a successful implementation, different departments (e.g., center of excellence for PM, audit data analytics) within the audit firms are involved in the implementation process. On top, the involvement of PM start-ups and software providers with additional guidance can be beneficial, as most tools are only offered as platforms without any developed audit program. Prepared and predefined analyses regarding a specific audit task that are consistent with the risk-based audit approach are necessary. Auditors in the center of excellence must map PM analyses to audit assertions and develop a PM audit program. This development is time-consuming. In the current state this phase is dynamic, and changes occur regularly. Therefore, documentation and use case descriptions of audit procedures must be adapted to meet the audit quality requirements. The implementation strategy needs to be coordinated within the audit teams. The project responsibility will be transferred after the initial phase from the audit data analytics teams to the audit engagement teams. So, audit engagement teams need to be composite with interdisciplinary team members for technical as well as business and accounting knowledge. All team members shall be trained in technical and business process fundamentals in advance.

6. Artifact evaluation

We verified and validated the overall usefulness of our developed *APM Framework* with quantitative evidence obtained from a survey with two Big Four audit firms and one Next Ten audit firm.⁷ We asked 19 professionals involved in the PM projects within their organizations to evaluate our developed artifact based on established DSR evaluation criteria ([Hevner et al., 2004](#); [March and Smith, 1995](#); [Sonnenberg and vom Brocke, 2012](#); [Venable et al., 2012](#)).

⁷ The whole evaluation instrument is attached in Appendix III.

Analyzing the responses of our participants, we identified that most of them (89 %) stated that they already utilize PM during an audit.⁸ Asking the participants for the most severe implementation barriers associated with PM (Table 3), further demonstrated the need for more guidance and understanding regarding how to implement PM to conduct audit procedures. Considering the top-three of the most severe implementation barriers indicates that 63 % of participants see that there is no understanding of PM among users, which indicates a lack of use by auditors outlined by 47 % of the participants. Furthermore, 47 % of participants highlighted that there also exist challenges for creating the data model.

Table 4 shows the results based on the questions asked about our developed *APM Framework* after demonstrating its functionality to the experts. The participants agreed that the *APM Framework* has an overall usefulness in overcoming existing implementation barriers for the audit profession and that it is easy to use. We asked the participants to agree or disagree with how well our developed *APM Framework* achieved the DSR evaluation criteria of overall usefulness to overcome existing implementation barriers and ease of use on a 7-point scale. The gathered evaluation results illustrate that the received assessments were statistically above the neutral midpoint of the scale, which indicates an overly positive evaluation of achieving the questioned DSR evaluation criteria (all p-values < 0.001, two-tailed) and the previously defined design objectives. Additionally, a great majority of the participants (95 %) highlighted, that they would use the *APM Framework* if they had access to it and 74 % of the experts think that the presented *APM Framework* will improve the performance of their PM-projects within the organization.

Overall, we believe that the feedback from two Big Four audit firms and one Next Ten audit firm, provides strong evidence toward the overall usefulness of the *APM Framework* to overcome existing implementation barriers associated with PM. Therefore, the framework achieves its major objectives and it can be defined as a verified and validated DSR artifact to solve existing real-world problems.

7. Limitations and future work

Although we rigorously developed and evaluated our *APM Framework* according to the DSR methodology of Peffers et al. (2007), and we managed to ensure that our central artifact can be defined as a relevant DSR contribution according to Gregor and Hevner (2013), the study at hand comes with several limitations providing opportunities for future research.

To develop the last level of abstraction of our *APM Framework* (process instance) and to demonstrate the framework to its target stakeholders, we chose to conduct expert interviews and selected our expert interviewees purposively from our professional network. To avoid bias, the experts were selected from different audit firms including Big Four (4 interviewees) and Next Ten (6 interviewees), large industrial companies (3 interviewees), and the academic arena (1 interviewee). As this process mitigates the problem of *unrepresentativeness* regarding our empirically gathered insights to a certain extent, this problem cannot be completely avoided with 14 interviewees out of our professional network. There may be resistance to some issues of subjectivity associated with the selection of our experts. Nevertheless, we tried to demonstrate the *APM Framework* to as many different experts from distinct audit firms and industrial companies as possible.

Although the evaluation of the *APM Framework* is based on established DSR evaluation criteria (Hevner et al., 2004; March and Smith, 1995; Sonnenberg and vom Brocke, 2012; Venable et al., 2012), the evaluation setting is rather simplistic. This is attributable to the innovative design and first development stage of the *APM Framework* presented in this study. However, measuring ease of use and overall usefulness to evaluate innovative artifacts is in alignment with other DSR studies in Accounting Information Systems (AIS), as shown in the study of Eulerich et al. (2022). Additionally, there are DSR studies in AIS that focus on some selected phases of DSR without the conduction of any evaluation, e.g., the study of Appelbaum and Nehmer (2017). We explicitly decided to provide a more simplistic evaluation setting to reach as many experts as possible, and thus to ensure some kind of representativeness. However, perceived ease of use and perceived usefulness represent defined constructs in the technology acceptance model (Davis, 1989). Therefore, subsequent studies could more empirically evaluate the *APM Framework* measuring these constructs based on the 14 questions provided by the technology acceptance model for each of the previously mentioned constructs.

Moreover, the developed *APM Framework* needs to be piloted in practice as an addition to our conducted evaluation to further evaluate its overall usefulness and implications of correctly implemented PM toward audit effectiveness and efficiency. Future studies in AIS can focus on the implementation of the *APM Framework* within real-world settings to analyze the resulting implications for the profession.

The measured lack of implementation of the use of PM can be analyzed from a different perspective instead of focusing on providing more guidance for the profession. For example, future research can focus more on the implementation barriers of the existing lack of

⁸ The fact that not everybody stated “yes” is attributable to some technical experts involved in our survey. These technical experts were more focused on data model creation and event log creation instead of the application of PM in audits. Nevertheless, as these technical experts were an essential part of PM projects in audit firms, we explicitly decided to include their views and responses in our evaluation.

use by auditors or challenges in creating the data model.

The scarce use of PM in auditing might also indicate that the costs (e.g., technology know-how, need for interdisciplinarity of the audit engagement team, license costs, etc.) of its implementation outweigh the benefits associated with the technology. This concern is also worth further investigation in future research.

Additionally, audit firms with clients who have few digitized business processes cause significant system discontinuities and thus mostly lead to mined process models with little value. In such cases, the relevant business processes must still be derived through interviews with the client.

8. Conclusion

PM is an emerging technology and is currently often used on a pilot basis within several audit firms or IAFs. The steps from piloting to widespread deployment of PM require mastering several challenges, such as technology know-how to utilize and implement PM and achieving the required interdisciplinarity within each audit engagement team to interpret, validate, and verify the output of the mined process analyses. With our research at hand, we go one step ahead, and show one possible implementation strategy to successfully implement PM for audit tasks.

We developed the *APM Framework* based on the combination of CRISP-DM and PM². We adapted this theoretically developed framework to auditing by implementing the relevant ISAs and IPPF AS and IPPF PS within each phase of the framework. Additionally, we gathered empirical data demonstrating the framework to 14 experts to further align our framework with practical and relevant insights to fill up a process instance view.

Overall, our framework provides detailed guidance about the implementation of PM in risk-based auditing for the entire audit profession. The quantitative evaluation with 19 experts out of two Big Four and one Next Ten audit firm yields statistically significant results and therefore indicates the overall usefulness of our developed artifact to overcome existing implementation barriers associated with PM. This ensures that the *APM Framework* can be defined as a DSR research contribution.

Nevertheless, this study comes with some limitations within its design settings according to the potential biases of interviewees, different requirements for varying sizes of audit firms, lack of pilot testing phases of the *APM Framework* in practice, and a merely simple evaluation design. Further limitations associated with the developed framework are particularly evident for smaller audit firms and can be addressed by subsequent PM studies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the participants of the 11th International Symposium on Accounting Information Systems (ISAIS 2023) for their helpful comments and recommendations.

APPENDIX I – QUESTIONS FOR PROBLEM IDENTIFICATION

Question 1: Can you briefly describe the current use of Process Mining in your organization?

Question 2: Which application areas (external auditing, internal auditing, forensics, consulting) do you think Process Mining is best suited to?

Question 3: What socio-technical factors influence the use of Process Mining in your organization?

Question 4: What challenges have you encountered when using Process Mining?

APPENDIX II – DATA COLLECTION TEMPLATE

Num.	Phase	Description	Empirical Insights
1	Business Process Understanding and Planning	1. Input Factors: Business processes of the firm 2. Output Factors: Business and process mining objectives and success criteria; information systems for data extraction 3. Activities: (1) Define the audit objectives; (2) Select the relevant business processes; (3) Build the audit team	
2	Event Data Understanding and Extraction	1. Input Factors: Business and process mining objectives and the underlying information systems for the execution of the selected business processes. 2. Output Factors: Event data without case notion and event classes. 3. Activities: (1) Identify and understand available data to achieve audit objectives; (2) Integrate different data sources from various information systems to define and extract event data; (3) Transfer existing process knowledge	
3	Data Model and Event Logs Creation	1. Input Factors: Event data 2. Output Factors: Data model and event logs as a basis for the modeling and mining phase of the framework 3. Activities: (1) Create data model to define relations between extracted data; (2) Create event logs by defining case notions and event classes based on event data	
4	Process Mining and Analysis	1. Input Factors: Data model and event logs 2. Output Factors: Fulfillment of the business and process mining objectives; performance and compliance findings of analyzed business processes 3. Activities: (1) Process discovery; (2) Conformance checking; (3) Enhancement and process analytics	
4.1	Process Discovery	Based on the previously extracted event logs, a discovery technique (alpha algorithm) is used to create a Petri net that represents the control flow of each event log.	
4.2	Conformance Checking	Conformance checking aims to identify inconsistencies between a process model (desired process sequence) and the actual process execution (execution log).	
4.3	Enhancement	Process enhancement focuses on the extension of an existing process model with further perspectives and information.	
4.4	Process Analytics	Process analytics refer to the use of data mining aspects to generate further insights and findings (decision trees, clustering methods).	
5	Audit Objective Evaluation	1. Input Factors: Process models, performance and compliance findings 2. Output Factors: Gathered audit evidence; improvement ideas and new business and process mining goals 3. Activities: (1) Assess the extent to which the audit objectives can be achieved with PM; (2) Verify and validate results; (3) Evaluate if further iterations between phases 1–4 are necessary	
6	Deployment, Maintenance, and Improvement	1. Input Factors: Results of the evaluation phase 2. Output Factors: Implemented process mining system; process modifications 3. Activities: (1) Deploy PM if audit objectives can be achieved; (2) Embed and scale PM into the existing audit methodology; (3) Implement monitoring and maintenance mechanisms to periodically review PM functionalities	
Other Notes			

Fig. A1. Data Collection Template

APPENDIX III – EVALUATION INSTRUMENT

Part I – Before the Presentation of the APM Framework.

Question 1: Do you already use PM in auditing?

Options: Yes or No.

Question 2: What are the biggest challenges of using PM in auditing?

Options: (1) Challenges in creating the data model; (2) Costs of implementation; (3) Inconsistent data basis; (4) Lack of understanding among users; (5) Lack of use by auditors; (6) No clear use cases within the audit department

Part II – After the presentation of the APM Framework

Rating: Evaluate the *APM Framework* shown (scales: –3 strongly disagree to 3 strongly agree) according to the following statements:

Statement 1: The *APM Framework* is useful.

Statement 2: The *APM Framework* is easy to use.

Question 1: Do you think the *APM Framework* shown will improve the performance of your PM projects?

Options: Yes or No.

Question 2: If you had access to the *APM Framework* shown and were working with PM, would you use it?

Options: Yes or No.

Data availability

The data that has been used is confidential.

REFERENCES

- Alles, M.G., Dai, J., Vasarhelyi, M.A., 2021. Reporting 4.0: Business reporting for the age of mass customization. *Journal of Emerging Technologies in Accounting*. 18 (1), 1–15. <https://doi.org/10.2308/jeta-10764>.
- Amani, F.A., Fadlalla, A.M., 2017. Data mining applications in accounting: A review of the literature and organizing framework. *International Journal of Accounting Information Systems*. 24, 32–58. <https://doi.org/10.1016/j.accinf.2016.12.004>.
- Appelbaum, D., Kogan, A., Vasarhelyi, M.A., 2017. Big data and analytics in the modern audit engagement: Research needs. *Auditing: A Journal of Practice & Theory*. 36 (4), 1–27. <https://doi.org/10.2308/ajpt-51684>.
- Appelbaum, D.A., Kogan, A., Vasarhelyi, M.A., 2018. Analytical procedures in external auditing: A comprehensive literature survey and framework for external audit analytics. *Journal of Accounting Literature*. 40 (1), 83–101. <https://doi.org/10.1016/j.acclit.2018.01.001>.
- Appelbaum, D., Nehmer, R.A., 2017. Using drones in internal and external audits: An exploratory framework. *Journal of Emerging Technologies in Accounting*. 14 (1), 99–113. <https://doi.org/10.2308/jeta-51704>.
- Austin, A.A., Carpenter, T.D., Christ, M.H., Nielson, C.S., 2021. The data analytics journey: Interactions among auditors, managers, regulation, and technology. *Contemporary Accounting Research*. 38 (3), 1888–1924. <https://doi.org/10.1111/1911-3846.12680>.
- Baader, G., Krcmar, H., 2018. Reducing false positives in fraud detection: Combining the red flag approach with process mining. *International Journal of Accounting Information Systems*. 31, 1–16. <https://doi.org/10.1016/j.accinf.2018.03.004>.
- Berger, S., Bitzer, M., Häckel, B., Voit, C., 2020. Approaching digital transformation – development of a multi-dimensional maturity model, in: *Proceedings of the 28th European Conference on Information Systems (ECIS 2020)*, vol. 181. AIS Electronic Library. https://aisel.aisnet.org/ecis2020_rp/181.
- Buchheit, S., Dzuránin, A.C., Hux, C., Riley, M.E., 2020. Data visualization in local accounting firms: Is slow technology adaption rational? *Current Issues in Auditing*. 14 (2), A15–A24. <https://doi.org/10.2308/cia-2019-501>.
- Burton, F.G., Summers, S.L., Wilks, T.J., Wood, D.A., 2021. Do we matter? Attention the general public, policymakers, and academics give to accounting research. *Issues in Accounting Education*. 36 (1), 1–22. <https://doi.org/10.2308/ISSUES-2020-017>.
- Celonis, 2017. Process mining and internal audit: A match made in heaven? <https://www.celonis.com/blog/process-mining-and-internal-audit-a-match-made-in-heaven/> (accessed 10 March 2023).
- Celonis, Fraunhofer Institute, 2022. Accelerating business transformation with process mining centers of excellence (CoEs): A study on the power of CoEs in achieving ROI. <https://publica.fraunhofer.de/entities/publication/94257ff7-3368-4d25-a114-95858e736c8e/details> (accessed 10 March 2023).
- Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., Wirth, R., 1999. CRISP-DM 1.0: Step-by-step data mining guide. <https://www.kde.cs.uni-kassel.de/wp-content/uploads/lehre/ws2012-13/kdd/files/CRISPWP-0800.pdf> (accessed 28 April 2024).
- Chiu, T., Brown-Liburd, H., Vasarhelyi, M.A., 2019. Performing tests of internal controls using process mining. *The CPA Journal*. 89 (6), 54–57.
- Chiu, T., Jans, M., 2019. Process mining of event logs: A case study evaluating internal control effectiveness. *Accounting Horizons*. 33 (3), 141–156. <https://doi.org/10.2308/acch-52458>.
- Chiu, T., Wang, Y., Vasarhelyi, M.A., 2020. The automation of financial statement fraud detection: A framework using process mining. *Journal of Forensic and Investigative Accounting*. 12 (1), 86–108.
- Christ, M.H., Emnett, S.A., Summers, S.L., Wood, D.A., 2021a. Prepare for takeoff: Improving asset measurement and audit quality with drone-enabled inventory audit procedures. *Review of Accounting Studies*. 26 (4), 1323–1343. <https://doi.org/10.1007/s11142-020-09574-5>.
- Christ, M.H., Eulerich, M., Krane, R., Wood, D.A., 2021b. New frontiers for internal audit research. *Accounting Perspectives*. 20 (4), 449–475. <https://doi.org/10.1111/1911-3838.12272>.
- Daglienè, L., Klovienė, L., 2019. Motivation to use big data and big data analytics in external auditing. *Managerial Auditing Journal*. 34 (7), 750–782. <https://doi.org/10.1108/MAJ-01-2018-1773>.
- Davenport, T. H., D'Ignazio, A., 2019. What process mining is, and why companies should do it. <https://hbr.org/2019/04/what-process-mining-is-and-why-companies-should-do-it> (accessed 10 March 2023).
- Davis, F.D., 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*. 13 (3), 319–340.
- Dilla, W.N., Raschke, R.L., 2015. Data visualization for fraud detection: Practice implications and a call for future research. *International Journal of Accounting Information Systems*. 16, 1–22. <https://doi.org/10.1016/j.accinf.2015.01.001>.
- Eilifsen, A., Kinsersdal, F., Messier, W.F., McKee, T.E., 2020. An exploratory study into the use of audit data analytics on audit engagements. *Accounting Horizons*. 34 (4), 75–103. <https://doi.org/10.2308/HORIZONS-19-121>.
- Eulerich, M., Huang, Q., Vasarhelyi, M. A., 2023. Using process mining as an assurance-tool in the three-lines-of-defense Model. SSRN Working Paper. Doi: 10.2139/ssrn.4494510.
- Eulerich, M., Pawlowski, J., Waddoups, N.J., Wood, D.A., 2022. A framework for using robotic process automation for audit tasks. *Contemporary Accounting Research*. 39 (1), 691–720. <https://doi.org/10.1111/1911-3846.12723>.
- EY, 2019. Process mining and its impact on BPM. <https://assets.ey.com/content/dam/ey-sites/ey-com/fi/fi/pdf/ey-process-mining-and-its-impact-on-bpm.pdf> (accessed 10 March 2023).
- Geerts, G.L., 2011. A design science research methodology and its application to accounting information systems research. *International Journal of Accounting Information Systems*. 12, 142–151. <https://doi.org/10.1016/j.accinf.2011.02.004>.

- Gehrke, N., 2010. Basic principles of financial process mining: A journey through financial data in accounting information systems. In: Proceedings of the 16th Americas Conference on Information Systems, vol. 289. AIS Electronic Library, Lima, Peru. <https://aisel.aisnet.org/amcis2010/289/>.
- Gepp, A., Linnenluecke, M.K., O'Neill, T.J., Smith, T., 2018. Big data techniques in auditing research and practice: Current trends and future opportunities. *Journal of Accounting Literature*. 40, 102–115. <https://doi.org/10.1016/j.acclit.2017.05.003>.
- Gregor, S., Hevner, A.R., 2013. Positioning and presenting design science research for maximum impact. *MIS Quarterly*. 37 (2), 337–355. <http://www.jstor.org/stable/43825912>.
- Grisold, T., Mendling, J., Otto, M., vom Brocke, J., 2021. Adoption, use and management of process mining in practice. *Business Process Management Journal*. 27 (2), 369–387. <https://doi.org/10.1108/BPMJ-03-2020-0112>.
- Hevner, A.R., March, S.T., Park, J., Ram, S., 2004. Design science in information systems research. *MIS Quarterly*. 28 (1), 75–105. <https://doi.org/10.2307/25148625>.
- IAASB, 2015a. International standard on auditing 260 (Revised): Communication with those charged with governance. <https://www.iaasb.org/publications/international-standard-auditing-isa-260-revised-communication-those-charged-governance-0> (accessed 29 April 2024).
- IAASB, 2015b. International standard on auditing 700 (Revised): Forming an opinion and reporting of financial statements. https://www.iaasb.org/_flysystem/azure-private/publications/files/ISA-700-Revised_8.pdf (accessed 29 April 2024).
- IAASB, 2015c. International standard on auditing 701: Communicating key audit matters in the independent auditor's report. https://www.iaasb.org/_flysystem/azure-private/publications/files/ISA-701_2.pdf (accessed 29 April 2024).
- IAASB, 2016. International standard on auditing 250 (Revised): Consideration of laws and regulations in an audit of financial statements. <https://www.iaasb.org/publications/isa-250-revised-consideration-laws-and-regulations-audit-financial-statements-13> (accessed 29 April 2024).
- IAASB, 2019. International standard on auditing 315 (Revised 2019): Identifying and assessing the risks of material misstatement. https://www.iaasb.org/_flysystem/azure-private/publications/files/ISA-315-Full-Standard-and-Conforming-Amendments-2019-.pdf (accessed 29 April 2024).
- IAASB, 2020. International standard on auditing 220 (Revised): Quality management for an audit of financial statements. https://www.iaasb.org/_flysystem/azure-private/publications/files/IAASB-International-Standard-Auditing-220-Revised.pdf (accessed 29 April 2024).
- IAASB, 2021. Technology frequently asked questions March 2021. <https://www.ifac.org/system/files/publications/files/IAASB-Automated-Tools-Techniques-FAQ.pdf> (accessed 10 March 2023).
- IFAC, 2009a. International standard on auditing 200: Overall objectives of the independent auditor and the conduct of an audit in accordance with international standards on auditing. https://www.ifac.org/_flysystem/azure-private/publications/files/A009%202012%20IAASB%20Handbook%20ISA%20200.pdf (accessed 29 April 2024).
- IFAC, 2009b. International standard on auditing 230: Audit documentation. https://www.ifac.org/_flysystem/azure-private/publications/files/A012%202012%20IAASB%20Handbook%20ISA%20230.pdf (accessed 29 April 2024).
- IFAC, 2009c. International standard on auditing 240: The auditor's responsibilities relating to fraud in an audit of financial statements. https://www.ifac.org/_flysystem/azure-private/publications/files/A012%202013%20IAASB%20Handbook%20ISA%20240.pdf (accessed 29 April 2024).
- IFAC, 2009d. International standard on auditing 300: Planning an audit of financial statements. https://www.ifac.org/_flysystem/azure-private/publications/files/A016%202013%20IAASB%20Handbook%20ISA%20300.pdf (accessed 29 April 2024).
- IFAC, 2009e. International standard on auditing 320: Materiality in planning and performing an audit. https://www.ifac.org/_flysystem/azure-private/publications/files/A019%202012%20IAASB%20Handbook%20ISA%20320.pdf (accessed 29 April 2024).
- IFAC, 2009f. International standard on auditing 330: The auditor's responses to assessed risks. https://www.ifac.org/_flysystem/azure-private/publications/files/A019%202013%20IAASB%20Handbook%20ISA%20330.pdf (accessed 29 April 2024).
- IFAC, 2009g. International standard on auditing 500: Audit evidence. https://www.ifac.org/_flysystem/azure-private/publications/files/A023%202012%20IAASB%20Handbook%20ISA%20500.pdf (accessed 29 April 2024).
- IFAC, 2009h. International standard on auditing 520: Analytical procedures. https://www.ifac.org/_flysystem/azure-private/publications/files/A026%202013%20IAASB%20Handbook%20ISA%20520.pdf (accessed 29 April 2024).
- IFAC, 2009i. International standard on auditing 530: Audit sampling. https://www.ifac.org/_flysystem/azure-private/publications/files/A027%202013%20IAASB%20Handbook%20ISA%20530.pdf (accessed 29 April 2024).
- IFAC, 2009j. International standard on auditing 620: Using the work of an auditor's expert. https://www.ifac.org/_flysystem/azure-private/publications/files/A035%202013%20IAASB%20Handbook%20ISA%20620.pdf (accessed 29 April 2024).
- IIA, 2016. International standards for the professional practice of internal auditing (standards). <https://www.theiia.org/globalassets/site/standards/mandatory-guidance/ippf/2017/ippf-standards-2017-english.pdf> (accessed 29 April 2024).
- IIA, 2017. International professional practices framework (IPPF), The IIA, Altamonte Springs, FL, USA.
- Issa, H., Sun, T., Vasarhelyi, M.A., 2016. Research ideas for artificial intelligence in auditing: The formalization of audit and workforce supplementation. *Journal of Emerging Technologies in Accounting*. 13 (2), 1–20. <https://doi.org/10.2308/jeta-10511>.
- Jans, M., 2019. Auditor choices during event log building for process mining. *Journal of Emerging Technologies in Accounting*. 16 (2), 59–67. <https://doi.org/10.2308/jeta-52496>.
- Jans, M., Alles, M.G., Vasarhelyi, M.A., 2013. The case for process mining in auditing: Sources of value added and areas of application. *International Journal of Accounting Information Systems*. 14 (1), 1–20. <https://doi.org/10.1016/j.accinf.2012.06.015>.
- Jans, M., Alles, M.G., Vasarhelyi, M.A., 2014. A field study on the use of process mining of event logs as an analytical procedure in auditing. *The Accounting Review*. 89 (5), 1751–1773. <https://doi.org/10.2308/accr-50807>.
- Jans, M., Eulerich, M., 2022. Process mining for financial auditing. In: van der Aalst, W.M.P., Carmona, J. (Eds.), *Process mining handbook*, vol. 448. Springer, Berlin, Germany, pp. 445–467. https://doi.org/10.1007/978-3-031-08848-3_15.
- Jans, M., Hosseinpour, M., 2019. How active learning and process mining can act as continuous auditing catalyst. *International Journal of Accounting Information Systems*. 32, 44–58. <https://doi.org/10.1016/j.accinf.2018.11.002>.
- Jans, M., van der Werf, J.M., Lybaert, N., Vanhoof, K., 2011. A business process mining application for internal transaction fraud mitigation. *Expert Systems with Applications*. 38, 13351–13359. <https://doi.org/10.1016/j.eswa.2011.04.159>.
- Kelleher, J. D., Tierney, B., 2018. *Data science*. The MIT Press, Cambridge, MA, USA.
- Kokina, J., Davenport, T.H., 2017. The emergence of artificial intelligence: How automation is changing auditing. *Journal of Emerging Technologies in Accounting*. 14 (1), 115–122. <https://doi.org/10.2308/jeta-51730>.
- Krieger, F., Drews, P., Velte, P., 2021. Explaining the (non-) adoption of advanced data analytics in auditing: A process theory. *International Journal of Accounting Information Systems*. 41, 100511. <https://doi.org/10.1016/j.accinf.2021.100511>.
- Leno, V., Polyvyanyy, A., Dumas, M., La Rosa, M., Maggi, F.M., 2021. Robotic process mining: Vision and challenges. *Business & Information Systems Engineering*. 63 (3), 301–314. <https://doi.org/10.1007/s12599-020-00641-4>.
- Lincoln, Y.S., Guba, E.G., 1985. *Naturalistic inquiry*. SAGE Publications Inc., Newbury Park, CA, USA.
- Lowe, D.J., Bierstaker, J.L., Janvri, D.J., Jenkins, J.G., 2018. Information technology in an audit context: Have the Big 4 lost their advantage? *Journal of Information Systems*. 32 (1), 87–107. <https://doi.org/10.2308/isys-51794>.
- Manita, R., Elommal, N., Buadier, P., Hikkerova, L., 2020. The digital transformation of external audit and its impact on corporate governance. *Technological Forecasting & Social Change*. 150, 119751. <https://doi.org/10.1016/j.techfore.2019.119751>.
- Mans, R., Reijers, H., Berends, H., Bandara, W., Prince, R., 2013. Business process mining success, in: *Proceedings of the 21st European Conference on Information Systems (ECIS 2013)*, vol. 89. AIS Electronic Library, Utrecht, Netherlands. https://aisel.aisnet.org/ecis2013_cr/89.
- March, S.T., Smith, G.F., 1995. Design and natural science research on information technology. *Decision Support Systems*. 15 (4), 251–266.

- Martin, N., Fischer, D.A., Kerpedzhiev, G.D., Goel, K., Leemans, S.J.J., Röglinger, M., van der Aalst, W.M.P., Dumas, M., La Rosa, M., Wynn, M.T., 2021. Opportunities and challenges for process mining in organizations: Results of a Delphi study. *Business & Information Systems Engineering*. 63 (5), 511–527. <https://doi.org/10.1007/s12599-021-00720-0>.
- Miles, M.B., Huberman, A.M., 1994. *Qualitative data Analytics: An expanded sourcebook*, second ed. SAGE Publications Inc., Thousand Oaks, CA, USA.
- Nickerson, R.C., Varshney, U., Muntermann, J., 2013. A method for taxonomy development and its application in information systems. *European Journal of Information Systems*. 22 (3), 336–359. <https://doi.org/10.1057/ejis.2012.26>.
- Park, G., van der Aalst, W.M.P., 2020. A general framework for action-oriented process mining. In: Del Río Ortega, A., Leopold, H., Santoro, F.M. (Eds.), *Business process management workshops: BPM 2020: Lecture notes in business information processing*, vol. 397. Springer, Cham, Switzerland, pp. 206–218. https://doi.org/10.1007/978-3-030-66498-5_16.
- Peffer, K., Tuunanen, T., Rothenberger, M., Chatterjee, S., 2007. A design science research methodology for information systems research. *Journal of Management Information Systems*. 24 (3), 45–77. <https://doi.org/10.2753/MIS0742-1222240302>.
- Peffer, K., Tuunanen, T., Niehaves, B., 2018. Design science research genres: Introduction to the special issue on exemplars and criteria for applicable design science research. *European Journal of Information Systems*. 27 (2), 129–139. <https://doi.org/10.1080/0960085X.2018.1458066>.
- Power, M.K., Gendron, Y., 2015. Qualitative research in auditing: A methodological roadmap. *Auditing: A Journal of Practice & Theory*. 34 (2), 147–165. <https://doi.org/10.2308/ajpt-10423>.
- Rowley, J., 2012. Conducting research interviews. *Management Research Review*. 35 (3/4), 260–271. <https://doi.org/10.1108/01409171211210154>.
- Rozinat, A., van der Aalst, W.M.P., 2008. Conformance checking of processes based on monitoring real behavior. *Information Systems*. 33 (1), 64–95. <https://doi.org/10.1016/j.is.2007.07.001>.
- Salijeni, G., Samsonova-Taddei, A., Turley, S., 2019. Big data and changes in audit technology: Contemplating a research agenda. *Accounting and Business Research*. 49 (1), 95–119. <https://doi.org/10.1080/00014788.2018.1459458>.
- Salijeni, G., Samsonova-Taddei, A., Turley, S., 2021. Understanding how big data technologies reconfigure the nature and organization of financial statement audits: A socio-material analysis. *European Accounting Review*. 30 (3), 531–555. <https://doi.org/10.1080/09638180.2021.1882320>.
- Sarno, R., Sinaga, F., Sungkono, K.R., 2020. Anomaly detection in business processes using process mining and fuzzy association rule learning. *Journal of Big Data*. 7, 5. <https://doi.org/10.1186/s40537-019-0277-1>.
- Schaefer, S.M., Alvesson, M., 2020. Epistemic attitudes and source critique in qualitative research. *Journal of Management Inquiry*. 29 (1), 33–45. <https://doi.org/10.1177/1056492617739155>.
- Simon, H.A., 1996. *The sciences of the artificial*, third ed. The MIT Press, Cambridge, MA, USA.
- Sonnenberg, C., vom Brocke, J., 2012. Evaluations in the science of the artificial – reconsidering the build-evaluate pattern in design science research. In: Peffer, K., Rothenberger, M., Kuechler, B. (Eds.), *Design science research in information systems: Advances in theory and practice: DESRIST 2012: Lecture notes in computer science*, vol. 7286. Springer, Berlin, Heidelberg, Germany, pp. 381–397. https://doi.org/10.1007/978-3-642-29863-9_28.
- Stephan, S., Lahann, J., Fetteke, P., 2021. A case study on the application of process mining in combination with journal entry tests for financial auditing. In: *Proceedings of the 54th Hawaii International Conference on System Sciences*. <http://hdl.handle.net/10125/71314>.
- Summers, S.L., Wood, D.A., 2017. An evaluation of the general versus specialist nature of top accounting journals. *Accounting Horizons*. 31 (2), 105–124. <https://doi.org/10.2308/acch-51712>.
- Sun, T.S., 2019. Applying deep learning to audit procedures: An illustrative framework. *Accounting Horizons*. 33 (3), 89–109. <https://doi.org/10.2308/acch-52455>.
- Thiede, M., Fuerstenau, D., Barquet, A.P.B., 2017. How is process mining technology used by organizations? A systematic literature review of empirical studies. *Business Process Management Journal*. 24 (4), 900–922. <https://doi.org/10.1108/BPMJ-06-2017-0148>.
- van der Aalst, W.M.P., et al., 2011. Process mining manifesto. In: Daniel, F., Barkaoui, K., Dustdar, S. (Eds.), *Business process management workshops: BPM 2011: Lecture notes in business information processing*, vol. 99. Springer, Berlin, Heidelberg, Germany, pp. 169–194. https://doi.org/10.1007/978-3-642-28108-2_19.
- van der Aalst, W.M.P., 2012. Process mining: Overview and opportunities. *ACM Transactions on Management Information Systems*. 3 (2), 1–17. <https://doi.org/10.1145/2229156.2229157>.
- van der Aalst, W.M.P., Weijters, T., Maruster, L., 2004. Workflow mining: Discovering process models from event logs. *IEEE Transactions on Knowledge and Data Engineering*. 16 (9), 1128–1142. <https://doi.org/10.1109/TKDE.2004.47>.
- van der Aalst, W.M.P., van Hee, K.M., van der Werf, J.M., Verdonk, M., 2010. Auditing 2.0: Using process mining to support tomorrow's auditor. *Computer*. 43 (3), 90–93. <https://doi.org/10.1109/MC.2010.61>.
- van der Aalst, W. M. P., 2016. *Process mining: Data science in action*, second ed. Springer, Berlin, Heidelberg, Germany. Doi: 10.1007/978-3-662-49851-4.
- van Eck, M. L., Lu X., Leemans, S. J. J., van der Aalst, W. M. P., 2015. PM²: A process mining project methodology, in: Zdravkovic, J., Kirikova, M., Johannesson, P. (Eds.), *Advanced information systems engineering: CAiSE 2015: Lecture notes in computer science*, vol. 9097. Springer, Cham, Switzerland, pp. 297–313. Doi: 10.1007/978-3-319-19069-3_19.
- Venable, J., Pries-Heje, J., Baskerville, R., 2012. A comprehensive framework for evaluation in design science research, in: Peffer, K., Rothenberger, M., Kuechler, B. (Eds.), *Design science research in information systems: Advances in theory and practice: DESRIST 2012: Lecture notes in computer science*, vol. 7286. Springer, Berlin, Heidelberg, Germany, pp. 423–438. Doi: 10.1007/978-3-642-29863-9_31.
- vom Brocke, J., Jans, M., Mendling, J., Reijers, H.A., 2021. A Five-Level Framework for Research on Process Mining. *Business & Information Systems Engineering*. 63 (5), 483–490. <https://doi.org/10.1007/s12599-021-00718-8>.
- Wang, Y., Chiu, T., Chiu, V., 2020. Redesigning business process to comply with the new revenue recognition standard using process mining. *Journal of Emerging Technologies in Accounting*. 17 (1), 149–163. <https://doi.org/10.2308/jeta-52663>.
- Waymire, G.B., 2012. Introduction for essays on the state of accounting scholarship. *Accounting Horizons* 26 (4), 817–819. <https://doi.org/10.2308/acch-50236>.
- Werner, M., 2016. Process model representation layers for financial audits. In: *Proceedings of the 49th Hawaii International Conference on System Sciences (HICSS)*. IEEE, pp. 5338–5347. <https://doi.org/10.1109/HICSS.2016.659>.
- Werner, M., 2017. Financial process mining – Accounting data structure dependent control flow inference. *International Journal of Accounting Information Systems*. 25, 57–80. <https://doi.org/10.1016/j.accinf.2017.03.004>.
- Werner, M., Gehrke, N., 2015. Multilevel process mining for financial audits. *IEEE Transactions on Services Computing*. 8 (6), 820–832. <https://doi.org/10.1109/TSC.2015.2457907>.
- Werner, M., Gehrke, N., 2019. Identifying the absence of effective internal controls: An alternative approach for internal control audits. *Journal of Information Systems*. 33 (2), 205–222. <https://doi.org/10.2308/isys-52112>.
- Werner, M., Wiese, M., Maas, A., 2021. Embedding process mining into financial statement audits. *International Journal of Accounting Information Systems*. 41, 100514. <https://doi.org/10.1016/j.accinf.2021.100514>.
- Werner, M., 2019. Materiality maps: Process mining data visualization for financial audits, in: *Proceedings of the 52nd Hawaii International Conference on System Sciences (HICSS 2019)*, pp. 1045–1054. <http://hdl.handle.net/10125/59544>.
- Zerbino, P., Aloini, D., Dulmin, R., Mininno, V., 2018. Process-mining-enabled audit of information systems: Methodology and an application. *Expert Systems With Applications*. 110, 80–92. <https://doi.org/10.1016/j.eswa.2018.05.030>.